

Kun Wang, PhD - CV

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Introduction

Computational biologist with a PhD in probability theory and mathematical statistics. Specializes in stochastic dynamical modeling and statistical inference for phylogeny-resolved single-cell data. Developed computational frameworks integrating lineage tracing and transcriptomics to reconstruct developmental dynamics and infer cell fate decisions. Led the development of PhyloVelo (*Nature Biotechnology*, 2023) and scPhyloX (*Cell Systems*, 2025), with additional contributions to mitochondrial lineage tracing analysis (*Genome Biology*, 2025). First author of a review article accepted in *Nature Reviews Genetics*.

Education

Xiamen University, Probability Theory and Mathematical Statistics

PhD

Xiamen, China

Sept 2022 – June 2026

- Thesis: Dynamical Modeling and Cell Fate Inference from Single-Cell Multimodal Lineage Tracing Data
- Advisor: Prof. Da Zhou

Xiamen University, Probability Theory and Mathematical Statistics

MS

Xiamen, China

Sept 2019 – June 2022

- Advisor: Prof. Da Zhou

Ocean University of China, Mathematics and Applied Mathematics

BS

Qingdao, China

Sept 2015 – June 2019

Experience

Shenzhen Institutes of Advanced Technology, Visiting Research Student

Shenzhen, China

Aug 2020 – June 2026

5 years 11 months

- Advisor: Dr. Zheng Hu
- Conducted research on lineage tracing and single-cell dynamics.
- Developed stochastic models integrating phylogeny and transcriptomics.

Research Contributions

PhyloVelo

June 2021 – June 2023

PhyloVelo infers cell state transitions using lineage-resolved single-cell data.

- Published in Nature Biotechnology (2023).

- Improves velocity estimation by incorporating lineage structure.
- Robustly maps developmental trajectories across diverse biological systems.

scPhyloX

Oct 2020 – Mar 2025

scPhyloX infers tissue growth dynamics from single-cell lineage trees

- Published in Cell Systems (2025).
- Quantifies diverse stem cell growth rates across different tissues.
- Detects Darwinian selection and expansion of advantageous tumor clones.

Skills

Programming: Python(Pytorch, Pyro, PyMC), R (Seurat), Matlab, Mathematica

Research Areas: Stochastic dynamical modeling, Variational inference, Single-cell lineage tracing, Transcriptomic analysis

Academic Service

- Peer reviewer, BMC Bioinformatics, 2026

Selected Publications

PhyloVelo Enhances Transcriptomic Velocity Field Mapping Using Monotonically Expressed Genes

May 2024

Kun Wang, Liangzhen Hou, et al.

[10.1038/s41587-023-01887-5](https://doi.org/10.1038/s41587-023-01887-5) (*Nature Biotechnology*)

Single-Cell Phylodynamic Inference of Stem Cell Differentiation and Tumor Evolution

May 2025

Kun Wang, Zhaolian Lu, et al.

[10.1016/j.cels.2025.101244](https://doi.org/10.1016/j.cels.2025.101244) (*Cell Systems*)

Computational approaches for multimodal lineage tracing

Apr 2026

Kun Wang, Xionglei He, Zheng Hu

(*Nature Reviews Genetics*, Accepted)

Clonal Expansion Dictates the Efficacy of Mitochondrial Lineage Tracing in Single Cells

Jan 2025

Xin Wang*, Kun Wang*, et al.

[10.1186/s13059-025-03540-7](https://doi.org/10.1186/s13059-025-03540-7) (*Genome Biology*)

Polyclonal-to-Monoclonal Transition in Colorectal Precancerous Evolution

Jan 2024

Zhaolian Lu, Shanlan Mo, et al.

[10.1038/s41586-024-08133-1](https://doi.org/10.1038/s41586-024-08133-1) (*Nature*)

Selected Honors

- Top 10 Bioinformatics Advances in China, 2023
- Best Oral Presentation, GPB Omics & Bioinformatics Frontiers Symposium, 2023

- BYD Scholarship, 2026
- SIAT Dean's Scholarship, 2023

Invited Talks

1. GPB Omics & Bioinformatics Frontiers Symposium 2023: "PhyloVelo enhances transcriptomic velocity field mapping using monotonically expressed genes" (Best Oral Presentation)
2. 2023 Genomics & AI: First Author Forum: "PhyloVelo enhances transcriptomic velocity field mapping using monotonically expressed genes"
3. 2025 Genomics & AI: First Author Forum: "Single-cell phylodynamic inference of stem cell differentiation and tumor evolution"
4. 2025 Quantitative Biology: Mathematical Modeling and Statistical Inference Workshop: "Stochastic modeling of cell fate dynamics in lineage-traced single-cell data"